

EXPERIMENT 16 – Customer Demographics Analysis

Question

Analyze customer age, gender, and income using R charts and tables.

Dataset

Customer ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

The supplied Customer Demographics dataset contains these three customer records.

R Code

```
# Experiment 16 - Customer Demographics

customer_id <- c(1, 2, 3)

age <- c(28, 35, 42)

gender <- c("Female", "Male", "Female")

income <- c(50000, 60000, 75000)

data <- data.frame(customer_id, age, gender, income)

# Question 1 - Age Bar Chart

barplot(age,

  names.arg = customer_id,

  main = "Customer Age Distribution",

  xlab = "Customer ID",

  ylab = "Age",

  col = "skyblue")

# Question 2 - Gender Pie Chart

gender_count <- table(gender)
```

```
pie(gender_count,  
    labels = paste(names(gender_count),  
                    gender_count),  
    main = "Customer Distribution by Gender")
```

Question 3 - Table

```
print(data)
```

Question 1

Create a bar chart to visualize the distribution of customer ages.

Output 1



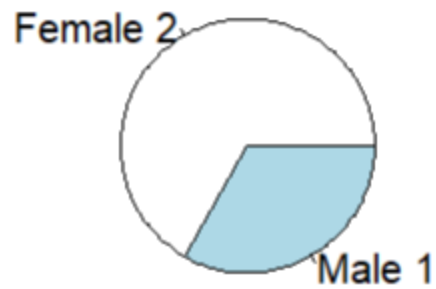
The bar chart displays customer ages of 28, 35, and 42 years.

Question 2

Generate a pie chart showing customer distribution by gender.

Output 2

Customer Distribution by Gender



The dataset contains:

- Female – 2 customers
- Male – 1 customer

Question 3

Display demographic information in a table.

Output 3

```
> # Question 3 - Table
> print(data)
  customer_id age gender income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
```